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Interlinked crises: the roles of food literacy, quality of life, and mental well-being in predicting adult food insecurity

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Abstract

Background Food insecurity and limited food literacy are important public health concerns associated with poorer psychological well-being and lower quality of life. This study aimed to examine the relationships between food insecurity, food literacy, quality of life, and mental well-being among adults in Türkiye.

Methods A cross-sectional online survey was conducted among adults aged 18–64 years in Türkiye ($n = 1,295$). Food insecurity was assessed using the USDA Food Security Survey Module (FSSM), food literacy was measured with the Short Food Literacy Questionnaire (SFLQ), quality of life was evaluated using the EUROHIS Quality of Life 8-Item Index (EUROHIS-QOL-8), and mental well-being was assessed with the Short Warwick–Edinburgh Mental Well-being Scale (SWEMWBS).

Results Food insecurity was prevalent in 44.1% of participants. Logistic regression analysis indicated that lower income ($p < 0.001$), lower education ($p < 0.001$), female gender ($p = 0.003$) and having more children ($p = 0.018$) were significantly associated with higher odds of food insecurity. The food-insecure group had lower food literacy ($p < 0.001$), quality of life ($p < 0.001$), mental well-being ($p < 0.001$), and number of employed individuals ($p < 0.01$). FSSM scores were negatively correlated with SFLQ ($p < 0.01$), EUROHIS-QOL-8 ($p < 0.01$), and SWEMWBS ($p < 0.01$). In linear regression analysis ($R^2 = 0.248$, $p < 0.001$), a one-year increase in age was associated with a 0.038-point decrease in FSSM total score. One-point increases in EUROHIS-QOL-8, SWEMWBS, and SFLQ scores were associated with decreases of 0.173, 0.042, and 0.064 points in FSSM total score, respectively. A one-unit increase in the number of employed individuals was associated with a 0.478-point decrease in FSSM total score.

Conclusions Food insecurity is associated with socioeconomic challenges, inadequate food literacy, lower quality of life, and reduced mental well-being. Policies integrating economic support with nutrition education, food literacy promotion, and mental health services may help address food insecurity and improve population well-being.

Keywords Food insecurity, Food literacy, Quality of life, Mental well-being, Socioeconomic status

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Text box 1. Contributions to the Literature

- This study demonstrates that food insecurity and inadequate food literacy are prevalent among adults in Türkiye, highlighting an important public health concern.
- It situates food insecurity within a broader psychosocial framework by linking it to food literacy, quality of life, and mental well-being.
- Quality of life is identified as a central determinant shaping food insecurity risk, extending current population health perspectives.
- The findings emphasize the need for integrated public health policies that address social and behavioral dimensions alongside economic factors.

Background

Food security exists when all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life. It is structured around four key dimensions: availability, access, utilization, and stability over time. If any of these dimensions is compromised, food insecurity may result [1]. Food insecurity can take many forms. Individuals may worry about running out of food before they can afford to buy more. They may also be unable to maintain a balanced diet, skip meals, or, in extreme cases, go without food for several days [2]. Food insecurity is also a major social determinant of health [3]. Individuals experiencing food insecurity are more vulnerable to chronic physical and mental health problems, as well as communicable and non-communicable diseases, compared with those who are food-secure [4].

Food literacy has received increasing attention as a factor associated with food insecurity [5]. It is defined as “a collection of interrelated knowledge, skills, and behaviors required to plan, manage, select, prepare, and eat food to meet needs and determine intake” [6]. Beyond knowledge and skills, food literacy enables individuals, households, and communities to maintain diet quality and make balanced food choices over time. Enhancing food literacy through education, skill-building, and support in navigating food-related policies and programs has been proposed as a strategy to promote healthy dietary practices and mitigate food insecurity [7]. Additionally, higher levels of food literacy have been linked to lower levels of food insecurity [8].

Food insecurity has been associated with quality of life and mental well-being outcomes [9]. Adults experiencing food insecurity report poorer life satisfaction, increased stress, and reduced psychological resilience [9, 10]. Although food insecurity and food literacy have been investigated in various populations [5, 11–14], there is still a lack of comprehensive research linking these factors to quality of life and mental well-being. Understanding these interrelationships is essential for developing effective strategies and public health policies

that promote healthier nutritional behaviors and psychological resilience. Therefore, the present study aimed to determine the prevalence of food insecurity among adults in Türkiye and to examine its associations with food literacy, quality of life, and mental well-being.

Methods**Study design and participants**

This descriptive cross-sectional study was conducted among adults aged 18–64 years. A total of 1295 individuals (531 men and 764 women) participated. Sample size was determined using an a priori power analysis conducted with G*Power version 3.1.9.7 [15]. An independent samples t-test (two-tailed) was used as the basis for the calculation. Assuming a medium effect size ($d = 0.40$), an alpha level of 0.05, and a statistical power of 0.90 ($1-\beta$), the minimum required sample size was calculated as 216 participants.

Participation was voluntary, and informed consent was obtained from all participants. Anonymity and confidentiality were ensured throughout the process.

Data collection

Data were collected online between February and April 2024 using a structured questionnaire administered via Google Forms. The survey link was disseminated via social media platforms such as WhatsApp, Instagram, and X. A snowball sampling approach was employed, where initial respondents were invited to share the survey with others in their networks. The questionnaire comprised six sections, including questions on sociodemographic characteristics, anthropometric measurements, food insecurity, food literacy, quality of life, and mental well-being.

Sociodemographic characteristics

The sociodemographic variables assessed included age, gender, marital status, educational level, number of children, number of household members, income level, employment status, and primary occupation. Additional items evaluated financial status (family income relative to the minimum wage; balance between income and expenses).

Anthropometric measurements

Self-reported weight and height were used to calculate Body Mass Index (BMI), defined as weight (kg) divided by height squared (m^2). BMI categories followed the World Health Organization (WHO) classification: underweight ($< 18.50 \text{ kg}/m^2$), normal weight ($18.50\text{--}24.99 \text{ kg}/m^2$), overweight ($25.00\text{--}29.99 \text{ kg}/m^2$), and obese ($\geq 30.00 \text{ kg}/m^2$) [16].

Food insecurity

Food insecurity status during the past 12 months was measured using the 10-item validated U.S. Department of Agriculture (USDA) Adult Food Security Survey Module (FSSM) [17]. Affirmative responses were summed to yield a total score between 0 and 10. Based on USDA guidelines, participants were categorized as high food security (score = 0), marginal food security (score = 1–2), low food security (score = 3–5), or very low food security (score = 6–10). As FSSM scores increase, levels of food security decrease, indicating higher levels of food insecurity. For statistical analyses, high and marginal food security were combined and defined as “food secure,” while low and very low food security were combined and defined as “food insecure [18]. In present study, Cronbach’s α coefficient was 0.880.

Food literacy

Food literacy was assessed using the Short Food Literacy Questionnaire (SFLQ) developed by Krause et al. [19]. The SFLQ consists of 12 items rated on four- or five-point Likert scales and evaluates functional, interactive, and critical food literacy within a unidimensional structure. Higher total scores indicate greater food literacy. In the original study, Cronbach’s α was 0.82. The Turkish adaptation and validation were conducted by Durmuş et al. [20], who reported a Cronbach’s α of 0.803. Participants scoring ≥ 31 points were categorized as having adequate to excellent food literacy, while those scoring < 31 points were classified as having inadequate to limited food literacy [21]. In the present study, Cronbach’s α was found to be 0.90.

Quality of life

Quality of life was evaluated using the World Health Organization Quality of Life Scale, Short Form (EURO-HIS-QOL-8) [22]. This eight-item unidimensional measure assesses overall quality of life, general health, energy, daily living activities, self-esteem, social relationships, finances, and home environment. Each item is rated on a five-point Likert scale, and higher scores represent better quality of life. The Turkish validation was conducted by Eser et al. [23], who reported a Cronbach’s α of 0.85. In the present study, Cronbach’s α was found to be 0.90.

Mental wellbeing

Mental well-being was measured using the Short Warwick-Edinburgh Mental Wellbeing Scale (SWEMWBS), a seven-item unidimensional short form of the WEMWBS developed by Stewart-Brown et al. [24]. Items are rated on a five-point Likert scale (1 = strongly disagree to 5 = strongly agree), yielding total scores ranging from 7 to 35, with higher scores indicating better mental well-being. The Turkish adaptation and validation were conducted

by Demirtaş and Baytemir [25], who reported Cronbach’s α values ranging from 0.84 to 0.86. In the present study, Cronbach’s α was found to be 0.92.

Statistical analysis

Data analyses were performed using the Statistical Package for the Social Sciences (SPSS), version 28.0. The normality of continuous variables was assessed using the Kolmogorov–Smirnov and Shapiro–Wilk tests. Categorical variables were summarized as frequencies and percentages, and group differences were examined using the chi-square test. Descriptive statistics for continuous variables were presented as number (n), mean ($\bar{X}$), and standard deviation (SD). Differences between two independent groups were evaluated using the independent samples t-test. Pearson’s correlation coefficients were calculated to assess the relationships among continuous variables. To identify psychosocial predictors of food insecurity, multivariable linear regression analysis was performed with FSSM total score as the dependent variable. Standardized beta coefficients (β) were calculated to compare the relative contribution of each predictor. The absolute values of the standardized β coefficients were plotted in a radar chart to visually display the relative strength of each predictor in the regression model. Additionally, binary logistic regression analysis was conducted to determine demographic factors associated with food insecurity, and results were reported as odds ratios (ORs) with 95% confidence intervals (CIs). Statistical significance was set at $p < 0.05$ for all analyses.

Results

Table 1 summarizes the descriptive characteristics of participants ($n = 1,295$). The mean age of individuals was 31.2 ± 11.60 years, ranging from 18 to 64 years. Among the participants, 59.0% were female and 41.0% were male. More than half of the respondents were single (53.6%), and most had a university degree or higher (66.9%). The mean household size was 5.0 ± 2.3 persons, with an average of 2.4 ± 1.28 children per household. The mean number of employed individuals per household was 1.4 ± 1.02 .

Regarding economic status, 11.2% of households reported an income below the minimum wage, 45.1% reported income at the level of the minimum wage, and only 43.7% earned above the minimum wage. Nearly half of the participants (44.3%) stated that their income was lower than their expenses. The unemployment rate among the participants was 56.4%, and the most common occupations were students (34.1%), civil servants (16.6%), and housewives (16.1%).

The mean FSSM score was 2.8 ± 3.20 , with 55.9% of individuals categorized as food-secure (FS) and 44.1% as food-insecure (FI). The mean SFLQ total score was 28.1 ± 9.15 , indicating that 60.0% of participants had

Table 1 Descriptive Characteristics of Participants ($n=1,295$)

	$\bar{X} \pm SD$ (Min-Max)
Age (years)	31.2 $\pm$ 11.60 (18.0–64.0)
Household size	5.0 $\pm$ 2.3 (0.0–14.0)
Number of children	2.4 $\pm$ 1.28 (1.0–7.0)
Number of employed individuals	1.4 $\pm$ 1.02 (0.0–9.0)
FSSM score	2.8 $\pm$ 3.20 (0.0–10.0)
SFLQ total score	28.1 $\pm$ 9.15 (5.0–49.0)
EUROHIS-QOL-8	26.8 $\pm$ 5.71 (8.0–40.0)
SWEMWBS	23.8 $\pm$ 6.08 (7.0–35.0)
BMI (kg/m^2)	24.9 $\pm$ 4.54 (14.5–44.1)
	n (%)
Gender	M: 531 (41.0%) F: 764 (59.0%)
Marital status	
Married	601 (46.4%)
Single	694 (53.6%)
Education level	
$\leq$ Secondary school	153 (11.8%)
High school	275 (21.2%)
$\geq$ University	867 (66.9%)
Household income	
< Minimum wage	145 (11.2%)
Minimum wage	584 (45.1%)
> Minimum wage	566 (43.7%)
Assessment of household income	
Income higher than expenses	504 (38.9%)
Income equals expenses	217 (16.8%)
Income lower than expenses	574 (44.3%)
Employment status	
Unemployed	731 (56.4%)
Employed	564 (43.6%)
Occupation type	
Civil servant	215 (16.6%)
Worker	58 (4.5%)
Private sector	191 (14.7%)
Self-employed	96 (7.4%)
Student	441 (34.1%)
Housewife	208 (16.1%)
Retired	52 (4.0%)
Other	34 (2.6%)
FSSM group	
High FS	468 (36.1%)
Marginal FS	256 (19.8%)
Low FS	276 (21.3%)
Very low FS	295 (22.8%)
SFLQ classification	
Inadequate-Limited FL	777 (60.0%)
Adequate-Excellent FL	518 (40.0%)
BMI classification	
Underweight	62 (4.8%)
Normal	623 (48.1%)

Table 1 (continued)

	$\bar{X} \pm SD$ (Min-Max)
Overweight	443 (34.2%)
Obese	167 (12.9%)

M Male, F Female, FL Food Literacy, FS Food security, FSSM Food Security Survey Module, SFLQ Short Food Literacy Questionnaire, EUROHIS-QOL-8 World Health Organization Quality of Life Scale, Short Form, SWEMWBS Warwick-Edinburgh Mental Wellbeing Scale, BMI Body Mass Index

inadequate-limited food literacy. The mean EUROHIS-QOL-8 and SWEMWBS scores were 26.8 ± 5.71 and 23.8 ± 6.08 , respectively, while the mean BMI was $24.9 \pm 4.54 \text{ kg}/\text{m}^2$, classifying 48.1% of participants as having normal weight, 34.2% as overweight, and 12.9% as obese.

Table 2 presents participant characteristics according to food insecurity status. Gender distribution did not differ significantly between FS and FI participants ($p=0.46$). However, marital status was significantly associated with food insecurity ($p=0.01$), with a higher proportion of unmarried individuals classified as food-insecure compared to married participants. No statistically significant differences were observed in education level ($p=0.65$). Employment-related variables exhibited strong associations with food insecurity. The proportion of employed individuals was significantly higher among food-secure participants (48.6%) than among food-insecure participants (37.1%) ($p<0.001$). Similarly, household income and income adequacy were significantly associated with food insecurity status ($p<0.001$). The proportion of participants with income above the minimum wage was notably higher in the FS group (54.1%) compared to the FI group (30.5%). In addition, households in the FS group more frequently reported income equal to or greater than expenses, whereas 57.3% of FI households reported income lower than expenses.

Occupation type was also related to food security ($p<0.001$). Food-secure individuals were more likely to be civil servants, self-employed, or private-sector employees, whereas students and housewives were more common among the food-insecure group. A significant association was also found between SFLQ classification and food insecurity ($p<0.001$), with inadequate-limited food literacy observed in 68.0% of food-insecure participants. No significant relationship was found between BMI classification and food insecurity status ($p=0.54$).

Factors associated with food insecurity are summarized in Table 3. Gender, total monthly income, education level, and the number of children in the household were significantly associated with food insecurity status ($p<0.05$). Males had lower odds of experiencing food insecurity compared with females (OR=0.56, 95% CI: 0.38–0.82, $p=0.003$). Each one-unit increase in monthly income was associated with a 74% reduction in the odds of food insecurity (OR=0.26, 95% CI: 0.15–0.45,

Table 2 Participant Characteristics by Food Insecurity Status ($n = 1,295$)

	FS ($n = 724$)	FI ($n = 571$)	<i>p</i>
Gender			
Male	290 (40.1%)	241 (42.2%)	0.46
Female	434 (59.9%)	330 (57.8%)	
Marital status			
Married	361 (49.9%)	240 (42.0%)	0.01
Single	363 (50.1%)	331 (58.0%)	
Education level			
≤Secondary school	87 (12.0%)	66 (11.6%)	0.65
High school	147 (20.3%)	128 (22.4%)	
≥University	490 (67.7%)	377 (66.0%)	
Employed status			
Employed	352 (48.6%)	212 (37.1%)	<0.001
Unemployed	372 (51.4%)	359 (62.9%)	
Household income			
< Minimum wage	56 (7.7%)	89 (15.6%)	<0.001
Minimum wage	276 (38.1%)	308 (53.9%)	
> Minimum wage	392 (54.1%)	174 (30.5%)	
Assessment of household income			
Income higher than expenses	321 (44.3%)	183 (32.0%)	<0.001
Income equals expenses	156 (21.5%)	61 (10.7%)	
Income lower than expenses	247 (34.1%)	327 (57.3%)	
Occupation type			
Civil servant	149 (20.6%)	66 (11.6%)	<0.001
Worker	32 (4.4%)	26 (4.6%)	
Private sector	101 (15.2%)	90 (15.8%)	
Self-employed	72 (9.9%)	24 (4.2%)	
Student	208 (28.7%)	233 (40.8%)	
Housewife	110 (15.2%)	98 (17.2%)	
Retired	29 (4.0%)	23 (4.0%)	
Other	23 (3.2%)	11 (1.9%)	
SFLQ classification			
Inadequate-Limited FL	389 (53.7%)	388 (68.0%)	<0.001
Adequate-Excellent FL	335 (46.3%)	183 (32.0%)	
BMI classification			
Underweight	31 (4.3%)	31 (5.4%)	0.54
Normal	347 (47.9%)	276 (48.3%)	
Overweight	257 (35.5%)	186 (32.6%)	
Obese	89 (12.3%)	78 (13.7%)	

FS Food security, FI Food insecurity, FL Food Literacy, SFLQ Short Food Literacy Questionnaire, BMI Body Mass Index

Table 3 Factors associated with the food insecurity status

Sociodemographic Characteristics	Odds Ratio (95% CI)	<i>p</i>
Gender	0.56 (0.38–0.82)	0.003
Total income (monthly)	0.26 (0.15–0.45)	<0.001
Education	0.39 (0.25–0.58)	<0.001
Household size	0.85 (0.56–1.26)	0.418
Number of children	2.01 (1.12–3.58)	0.018

Table 4 Food Literacy, Quality of Life, Mental Well-Being, and BMI According to Food Insecurity Status

Age	FS ($n = 724$)	FI ($n = 571$)	<i>p</i>
	31.7 ± 11.64	30.6 ± 11.52	0.08
FSSM total	0.4 ± 0.73	5.9 ± 2.47	<0.01
SFLQ total score	29.9 ± 8.86	26.1 ± 9.12	<0.01
EUROHIS-QOL-8	28.6 ± 5.24	24.4 ± 5.43	<0.01
SWEMWBS	25.3 ± 5.48	22.0 ± 6.30	<0.01
BMI	25.0 ± 4.41	24.9 ± 4.71	0.60
Household size	3.7 ± 1.38	3.8 ± 1.47	0.58
Number of employed individuals	1.6 ± 0.78	1.4 ± 0.75	<0.01

FS Food security, FI Food insecurity, FSSM Food Security Survey Module, SFLQ Short Food Literacy Questionnaire, EUROHIS-QOL-8 World Health Organization Quality of Life Scale, Short Form, SWEMWBS Warwick-Edinburgh Mental Wellbeing Scale, BMI Body Mass Index

$p < 0.001$). Higher educational attainment also demonstrated a protective effect (OR = 0.39, 95% CI: 0.25–0.58, $p < 0.001$). In contrast, the number of children in the household was positively associated with food insecurity (OR = 2.01, 95% CI: 1.12–3.58, $p = 0.018$), indicating that households with more dependent children were more vulnerable. The household size was not significantly associated with food insecurity ($p > 0.05$).

As shown in Table 4, no statistically significant differences were observed in age ($p = 0.08$), BMI ($p = 0.60$), or household size ($p = 0.58$) between participants based on food insecurity status. However, the number of employed individuals per household was significantly lower in the FI group ($p < 0.01$). Additionally, the FI group had significantly lower SFLQ, EUROHIS-QOL-8, and SWEMWBS scores than the FS group ($p < 0.01$ for all), whereas the FSSM score was significantly higher in the FI group ($p < 0.01$).

As shown in Table 5, significant correlations were observed among food insecurity, food literacy, quality of life, and mental well-being. FSSM scores were negatively correlated with SFLQ ($r = -0.296$, $p < 0.01$), EUROHIS-QOL-8 ($r = -0.426$, $p < 0.01$), and SWEMWBS ($r = -0.367$, $p < 0.01$), indicating that higher levels of food insecurity were associated with lower food literacy, poorer quality of life, and reduced mental well-being. SFLQ scores were positively correlated with both EUROHIS-QOL-8 ($r = 0.279$, $p < 0.01$) and SWEMWBS ($r = 0.281$, $p < 0.01$), suggesting that individuals with higher food literacy also tended to have better quality of life and psychological well-being. A strong positive correlation was found between EUROHIS-QOL-8 and SWEMWBS scores ($r = 0.697$, $p < 0.01$), highlighting the close association between general life satisfaction and mental well-being. In contrast, BMI was not significantly correlated with FSSM, EUROHIS-QOL-8, or SWEMWBS, indicating that body mass index was not directly associated with food insecurity, quality of life, or mental health in this sample. Age showed weak but significant

Table 5 Correlation Matrix of Study Variables

	1	2	3	4	5	6	7	8
1. FSSM	1							
2. SFLQ	-0.296**	1						
3. EUROHIS-QOL-8	-0.426**	0.279**	1					
4. SWEMWBS	-0.367**	0.281**	0.697**	1				
5. BMI	-0.027	-0.108**	-0.035	0.039	1			
6. Age	-0.119**	-0.092**	-0.016	0.113*	0.496**	1		
7. Household size	0.087**	-0.075**	-0.094**	-0.123**	-0.108**	-0.154**	1	
8. Number of employed individuals	-0.124**	0.137**	0.055*	0.011	-0.127**	-0.184**	0.234**	1

FSSM Food Security Survey Module, SFLQ Short Food Literacy Questionnaire, EUROHIS-QOL-8 World Health Organization Quality of Life Scale, Short Form, SWEMWBS Warwick-Edinburgh Mental Wellbeing Scale, BMI Body Mass Index

*Correlation is significant at the 0.05 level, **Correlation is significant at the 0.01 level

Table 6 Linear Regression Analysis of FSSM Total Score and Associated Factors

FSSM	β coefficient (95%CI)	Stan- dard- ized Beta	SE	t	p
Age	-0.038 (-0.051; -0.024)	-0.136	0.007	-5.433	<0.001
EUROHIS-QOL-8	-0.173 (-0.210; -0.135)	-0.307	0.019	-9.032	<0.001
SWEMWBS	-0.042 (-0.078; -0.007)	-0.080	0.018	-2.336	0.020
SFLQ	-0.064 (-0.081; -0.046)	-0.181	0.009	-7.022	<0.001
Number of employed individuals	-0.478 (-0.685; -0.272)	-0.116	0.105	-4.541	<0.001
Household size	0.092 (-0.020; 0.204)	0.041	0.057	1.606	0.109

FSSM Food Security Survey Module, SFLQ Short Food Literacy Questionnaire, EUROHIS-QOL-8, World Health Organization Quality of Life Scale, Short Form, SWEMWBS Warwick-Edinburgh Mental Wellbeing Scale

negative correlations with FSSM ($r = -0.119$, $p < 0.01$) and SFLQ ($r = -0.092$, $p < 0.01$). Household size was positively correlated with FSSM ($r = 0.087$, $p < 0.01$) but negatively with both EUROHIS-QOL-8 and SWEMWBS ($p < 0.01$). Additionally, the number of employed individuals was negatively correlated with FSSM ($r = -0.124$, $p < 0.01$) and positively correlated with SFLQ ($r = 0.137$, $p < 0.01$), indicating that employment contributed to better food insecurity and literacy outcomes.

Results of the linear regression analysis examining factors associated with the FSSM total score are presented in Table 6. The overall model was statistically significant ($R^2 = 0.248$, $p < 0.001$). A one-unit increase in age was associated with a 0.038-point decrease in the FSSM total score. Similarly, one-unit increases in EUROHIS-QOL-8, SWEMWBS, and SFLQ scores were associated with decreases of 0.173, 0.042, and 0.064 points in the FSSM total score, respectively. Additionally, a one-unit increase in the number of employed individuals was associated with a 0.478-point decrease in the FSSM total score.

Household size was not significantly associated with the FSSM total score ($p = 0.109$).

Discussion

This study aimed to assess the relationships between food insecurity, food literacy, quality of life, and mental well-being among adults in Türkiye. The results showed that nearly half of the participants (44.1%) experienced food insecurity. Food insecurity was more prevalent among females, unmarried individuals, those with lower education and income levels, and households with a higher number of children. Moreover, inadequate food literacy, lower quality of life, and poorer mental well-being were significantly associated with food insecurity.

Food insecurity remains a critical public health nutrition concern in both developed and developing countries. According to the 2024 *State of Food Security and Nutrition in the World* report, an estimated 2.3 billion people worldwide experienced moderate or severe food insecurity in 2024, corresponding to 28.0% of the global population [26]. Although there is no nationally representative estimate of food insecurity in Türkiye, studies conducted in different population groups report substantial prevalence rates, including 42.7% among adults [27], 21.6% among women of reproductive age [28], 68% among university students [29], and 30% among older adults [30]. In the present study, the prevalence of food insecurity was 44.1%. Variations across studies may be attributed to differences in sociodemographic characteristics of the study samples, as well as differences in measurement tools and classification criteria used to assess food insecurity. The prevalence observed in the present study may partly reflect Türkiye's prolonged period of high inflation, declining purchasing power, and increasing living costs [31].

Food insecurity is influenced by several sociodemographic factors, including gender, marital status, number of children, education level, and income [32, 33]. In this study, lower household income and a lower number of employed household members were associated with food insecurity, which is consistent with evidence showing

that limited financial resources reduce the ability to acquire food [34]. Women were also more affected than men, and gender-based inequalities in access to education, employment, and productive resources contribute to increasing the risk of poverty and inadequate food access among women and girls [35–37]. Marital status was another important factor, as unmarried individuals showed higher food insecurity, likely due to fewer opportunities for income sharing and resource pooling [4]. The number of children in the household also increased risk, since more dependents create greater financial pressure and caregiving responsibilities [38, 39]. Lower education level was also associated with higher vulnerability, consistent with evidence showing that individuals with less education and fewer food-related skills face greater challenges in securing and managing food resources [40, 41].

Food insecurity not only reflects economic hardship but also poses serious health risks. Studies have linked it to lower nutrient intake, poor sleep quality, higher rates of depression, and a greater prevalence of metabolic disorders such as type 2 diabetes and hypertension [2, 39]. Paradoxically, food insecurity has also been associated with overweight and obesity, particularly among women, due to reliance on inexpensive, calorie-dense, and nutrient-poor foods [42, 43]. However, in the present study, no significant differences in BMI were observed between food-secure and food-insecure participants. Differences in food insecurity measurement and classification, as well as energy-related factors such as physical activity, eating behaviors, gender, and age in the study population, may explain the absence of an association [44].

Food insecurity and food literacy showed a significant negative association in the present study. This finding is consistent with previous research. Begley et al. [5] reported that lower engagement in food literacy-related behaviors, such as planning, management, shopping, preparation, and cooking, was significantly associated with food insecurity. Similarly, Jomaa et al. [14] reported that food insecurity was associated with lower participation in food literacy practices, such as selecting, purchasing, and preparing healthy and nutritionally balanced foods. Limited food literacy may limit individuals' ability to plan, choose, and prepare affordable meals, thereby increasing their vulnerability to food insecurity, particularly under economic constraints [5, 45]. Additionally, in this study, inadequate food literacy was identified in 60.0% of participants, a proportion higher than those reported in earlier research, with Morgan et al. [46] observing inadequate food literacy in about 38–40% of university students and Hanbazaza and Alaklabi [13] reporting rates of 30–35% among adults. This difference may be related to the sociodemographic characteristics of the study population and to variations in how food literacy is assessed and classified across studies.

Consistent with previous research [47–49], higher levels of food insecurity in the present study were significantly associated with lower quality of life scores. Possible pathways that may help explain the association between food insecurity and lower quality of life include poor diet quality, limited access to healthy foods, financial difficulties, and increased stress. Together, these factors may contribute to reduced social participation, increased daily strain and feelings of insecurity, and lower perceived quality of life [50–52]. These findings highlight that food insecurity extends beyond material deprivation and is closely related to perceived quality of life through nutritional and social mechanisms.

Food insecurity is related to poorer mental health and well-being outcomes [53]. In this study, food insecurity showed a negative correlation with mental well-being, and a 1-unit increase in SWEMWBS was associated with a 0.042-point decrease in the FSSM total score. Evidence from analyses across 160 countries similarly demonstrates that food insecurity is strongly associated with reduced well-being and increased mental health symptoms, independent of income and other structural factors [54]. A recent meta-analysis further indicated that food insecurity significantly increases the risk of stress and depression [55], supporting our finding that individuals experiencing food insecurity had lower mental well-being. A recent study conducted in Türkiye reported that individuals experiencing food insecurity had significantly lower Flourish Index scores, indicating poorer overall well-being. The same study further highlighted that the adverse effects of food inflation were particularly pronounced among food-insecure groups, emphasizing the role of the current economic context in shaping well-being outcomes [56]. Several mechanisms may help explain these associations. Chronic uncertainty associated with food insecurity may trigger physiological stress responses, including activation of the hypothalamic–pituitary–adrenal axis, which can contribute to elevated cortisol levels and depressive symptoms [10]. Food insecurity also carries emotional and social consequences—such as shame, guilt, and social isolation—that further intensify psychological distress and undermine individuals' coping abilities [55, 57]. Together, these mechanisms illustrate how food insecurity adversely affects mental health and well-being and highlight the need for integrated interventions that strengthen psychological resilience alongside efforts to improve food security.

Limitations and strengths

There are several limitations to this study. First, the cross-sectional design allows for identifying associations between food insecurity, food literacy, quality of life, and mental well-being; however, it does not permit causal inference. Second, because data were collected online

through voluntary participation, the sample may not be representative of the broader adult population and may be biased toward individuals with higher education levels or more reliable internet access. Third, body weight and height were self-reported rather than objectively measured, which may have introduced measurement error and affected the accuracy of BMI classification. Despite these limitations, this study's strengths include the assessment of multiple related domains—food insecurity, food literacy, quality of life, and mental well-being—which allows for a more comprehensive understanding of their interactions in adults, while the large sample size strengthens the validity of the analyses and the use of validated instruments enhances the reliability of the findings.

Conclusions

This study showed that food insecurity was prevalent among adults and was significantly associated with lower food literacy, poorer quality of life, and reduced mental well-being. Regression analyses indicated that gender, education level, income status, and number of children were significantly related to food insecurity. Together, these findings highlight the multidimensional nature of food insecurity and underscore the importance of integrated public health strategies that combine nutrition education, social and economic support, and mental health promotion to improve overall well-being and resilience in the context of food insecurity.

Abbreviations

BMI	Body Mass Index
EUROHIS-QOL-8	European Health Interview Survey – World Health Organization Quality of Life 8-Item Index
FI	Food Insecurity
FS	Food Security
FSSM	Food Security Survey Module
SFLQ	Short Food Literacy Questionnaire
SPSS	Statistical Package for the Social Sciences
SWEMWBS	Short Warwick–Edinburgh Mental Well-being Scale
USDA	United States Department of Agriculture
WHO	World Health Organization

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Authors' contributions

KE: design, conceptualization, methodology, writing, review, and editing. FA: design, analysis, methodology, writing, review, and editing. EE: methodology, writing, review, and editing. The authors have approved the final version submitted.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Ethics Committee for Social and Human Sciences Research of Tokat Gaziosmanpaşa University (Decision No: 02.18; Date: 30.01.2024). The study adhered to the principles of the Declaration of Helsinki. All participants were informed about the aims of the study, and electronic informed consent was obtained prior to participation. Participation was voluntary, and respondents could withdraw at any time without consequence.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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